ELLIOT MYUNGHOON KIM

kim.elliot@northeastern.edu | linkedin.com/in/elliotk19 | Google Scholar | (+1) 607-280-8688

EDUCATION

Northeastern University — **Boston, MA**
PhD in Computer Science — Sept 2026 - Present

Cornell University — **Ithaca, NY**
Master of Engineering in Computer Science — Jan 2025 – Dec 2025
- **GPA**: 3.98

B.A. in Computer Science, B.A. in Information Science (Concentration: Data Science) — Aug 2019 – May 2025
- **GPA**: 3.91, Dean's List

PUBLICATIONS

[1] Ting, Y.-S., **Kim, E. M.** "The Payne Zero Project I: Stellar Spectra from Physical Models in Seconds." The Open Journal of Astrophysics, 9, 2026. [Link]

[2] Ting, Y.-S., Guhathakurta, P., Kirby, E. N., **Kim, E. M.** "Weak-CN Stars Are Ordinary Cool Red Supergiants." arXiv preprint arXiv:2605.30733, 2026. [Link]

[3] **Kim, E.**, Ting, Y.-S. "pyKurucz: A Pure Python Reimplementation of Kurucz ATLAS12 and SYNTHE for Stellar Spectrum Synthesis." arXiv preprint arXiv:2603.11693, 2026. [Link]

[4] **Kim, E.***, Garg, A.*, Peng, K.*, Garg, N. "Correlated Errors in Large Language Models." International Conference on Machine Learning (ICML), vol. 267, 2025, pp. 30038–30066. [Link]

RESEARCH EXPERIENCE

Rad Lab — **Boston, MA**
Research Assistant — Prof. Tina Eliassi-Rad — Sept 2026 – Present
- Leading an AI-as-co-scientist study comparing how different representations of large, dense graphs (up to 7.6M edges) affect LLM analysis of 172 gene co-expression networks.

UIUC ScaleML Lab — **Remote**
Research Assistant — Prof. Tong Zhang — Aug 2025 – May 2026
- Led a case study revealing performance degradation when using built-in web search tools of state-of-the-art LLMs on a challenging multi-modal physics benchmark.
- Built benchmark components spanning 217 geometry problems evaluated with AlphaGeometry's symbolic proof engine and 53 scientific-computing tasks evaluating LLM-generated Fortran against reference outputs.

Cornell Bowers Undergraduate Research Experience (BURE REU) — **Ithaca, NY**
Research Assistant — Prof. Nikhil Garg — June 2024 – Dec 2025
- Designed and implemented a cross-model evaluation showing that LLM errors are highly correlated, especially among newer, more accurate models.
- Across 1,800 resume–job pairs and 20 LLMs, showed that correlated errors can systematically exclude the same applicants in hiring pipelines.

Cornell Nexus Scholars Program — **Ithaca, NY**
Research Assistant – Prof. David Strang — May 2023 – Jul 2023
- Engineered an automated data pipeline, using Selenium and ETL processes to collect 65k NYT obituaries from Lexis Nexis into MongoDB; integrated Google Trends data to analyze search interest trends.

RESEARCH SOFTWARE

pyKurucz [Link] — **Remote**
Lead Developer — Prof. Yuan-Sen Ting — Nov 2025 – June 2026

- Developed and validated a modular Python reimplementation of Kurucz ATLAS12 and SYNTHE, legacy Fortran stellar-atmosphere codes, using coding agents [3].
- Enabled stellar spectrum experiments and integration with machine learning frameworks, supporting follow-up research on weak-CN stars [2].
- pyKurucz served as the foundation for Payne Zero [1], a GPU-native framework that reduced stellar spectrum synthesis time from tens of minutes to seconds.

PROFESSIONAL EXPERIENCE

Commure **Mountain View, CA**

Software Engineer – Revenue Cycle Management (RCM) Core Backend Feb 2026 – July 2026

Software Engineer Intern – Revenue Cycle Management (RCM) Core Backend June 2025 – Aug 2025

- Built an on-call Slack agent that aggregates the context engineers use to triage pages and auto-responds to tagged requests, cutting on-call overhead by 80%.
- Built an eligibility enrichment pipeline that resolves a claim's eligibility details from the patient's eligibility response before submission, resolving 65% of eligibility-related denials, the largest denial bucket.

Republic of Korea Army, 2nd Armored Brigade Air-Defense Unit **Paju-Si, Korea**

Sergeant, Command Control Communication Computer Intelligence (C4I) Operator Jul 2020 – Jan 2022

- Managed a group of 8 privates in the Air-Defense Operations Center utilizing radar & satellite warning systems.

TEACHING EXPERIENCE

- Head TA, INFO 3950: Data Analytics for Information, Cornell University, 2025
- TA, INFO 2950: Introduction to Data Science, Cornell University, 2023–2024
- Course Consultant, CS 1110: Introduction to Computing Using Python, Cornell University, 2022–2023

HONORS & AWARDS

- Cornell BURE REU Fellowship, 2024
- Cornell Nexus Scholars Program Fellowship, 2023
- Big Red Hacks Innovation Track Prize Winner, 2022

SKILLS

- **Languages:** English (Native), Korean (Native), Mandarin (Intermediate)
- **Programming Languages:** Python, R, Java, OCaml, Stata, Javascript (React), SQL, HTML, CSS
- **ML & Data:** PyTorch, Scikit-learn, Hugging Face, pandas